

LOVON: Legged Open Vocabulary Object Navigator

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Abstract—Object navigation in open-world environments remains a formidable and pervasive challenge for robotic systems, particularly when it comes to executing long-horizon tasks that require both open-world object detection and high-level task planning. Traditional methods often struggle to integrate these components effectively, and this limits their capability to deal with complex, long-range navigation missions. In this paper, we propose LOVON, a novel framework that integrates large language models (LLMs) for hierarchical task planning with open-vocabulary visual detection models, tailored for effective long-range object navigation in dynamic, unstructured environments. To tackle real-world challenges including visual jittering, blind zones, and temporary target loss, we design dedicated solutions such as Laplacian Variance Filter for visual stabilization. We also develop a functional execution logic for the robot that guarantees LOVON’s capabilities in autonomous navigation, task adaptation, and robust task completion. Extensive evaluations demonstrate the successful completion of long-sequence tasks involving real-time detection, search, and navigation toward open-vocabulary dynamic targets. Furthermore, real-world experiments across different legged robots (Unitree Go2, B2, and H1-2) showcase the compatibility and appealing plug-and-play feature of LOVON. Our code is available at <https://github.com/DaojiePENG/LOVON>.

Index Terms—Legged Robots, Open-Vocabulary Navigation, Robotic Autonomy

I. INTRODUCTION

IN recent years, large language models (LLMs) [1] and vision models [2]–[5] have achieved revolutionary breakthroughs in the field of artificial intelligence. LLMs have significantly improved their capabilities in understanding and planning long-horizon tasks, enabling them to deeply comprehend complex contexts and generate efficient execution strategies, which brings new possibilities to task planning. Meanwhile, advances in open-vocabulary visual detection have empowered vision models to recognize and understand a diverse range of objects beyond predefined categories, greatly enhancing the adaptability of machine vision systems in scene understanding and object recognition. These leaps in

perception and cognition lay a solid foundation for addressing complex long-horizon tasks in robotics.

Legged robots have evolved over decades and now demonstrate outstanding mobility in complex terrains. Their unique structural design and motion control allow them to adapt to various rugged environments, exhibiting terrain adaptability far beyond that of traditional wheeled robots. However, most current research focuses on optimizing single tasks, such as walking, jumping, climbing, and short-range navigation, lacking comprehensive consideration of complex long-horizon missions. It is acknowledged that the potential of legged robots to perform long-horizon tasks in open environments has not been fully explored, and that integrating advanced language and vision understanding with legged robot mobility is a key breakthrough for real-world applications.

In this paper, we propose LOVON, an innovative operating system that integrates the task planning capabilities of LLMs, the perception abilities of open-vocabulary visual detection, and a language-to-motion model (L2MM) for precise motion prediction, as shown in Fig. 1. The LOVON system addresses real-world challenges such as visual jitter caused by robot motion. In particular, we design a Laplacian Variance Filtering technique to effectively mitigate visual instability during robot movement, ensuring the accuracy and continuity of object detection. We also propose the functional execution logic for robust task completion. Experiments on simulation benchmark Gym-Unreal [6] demonstrate the superior results of LOVON. In addition, we conduct extensive experiments on multiple legged platforms (Unitree Go2, B2, and H1-2), successfully accomplishing long-horizon tasks involving real-time detection, search, and navigation toward open-vocabulary dynamic targets. To the best of our knowledge, LOVON is the first operational system to achieve such complex capabilities in unstructured environments. The main contributions of this work are summarized as follows:

- We propose LOVON, a unified framework that integrates LLMs, open-vocabulary visual detection, and a language-to-motion model, enabling the planning and execution of complex open-world long-horizon navigation tasks.
- We develop a Laplacian Variance Filtering method to resolve dynamic blurring issues and improve the robustness of the system. Also, we introduce a robot execution logic that ensures adaptability to various environments.
- We conduct comprehensive validation through simulations and real-world experiments across a variety of legged robot platforms. The results demonstrate that our system successfully performs open-vocabulary object search and navigation tasks in unstructured scenes.

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Fig. 1. Object navigation of legged robots in diverse open-world scenarios. These scenarios include executing long-horizon tasks with multiple subgoals (e.g., sequential navigation to multiple targets), achieving dynamic tracking of moving objects (e.g., following a person or animal), adapting to multi-configurations across different legged robot platforms (e.g., Unitree Go2, B2, H1-2), implementing searching for lost objects when targets are temporarily out of view, realizing retargeting of objects to switch between different task targets, and supporting open-vocabulary recognition to identify diverse objects beyond predefined categories—all of which reflect the comprehensive practical performance of LOVON in real experimental settings.

II. RELATED WORKS

A. Large Language Models for Robotic Task Planning

LLMs, such as GPT [1], PaLM [7], [8], and LLaMA [9], have demonstrated remarkable capabilities in natural language understanding, reasoning, and task decomposition. In the context of robotics, LLMs have been increasingly adopted for high-level task planning, instruction following, and semantic reasoning. Recent works leverage LLMs to interpret human instructions, generate action sequences, and adapt plans to dynamic environments, enabling robots to perform complex, long-horizon tasks with greater autonomy and flexibility. For example, SayCan [10] integrates LLMs with robotic affordance models to map language instructions to executable actions, while Code as Policies [11] utilizes LLMs to generate code for robot controllers directly from natural language descriptions. Despite these advances, challenges remain in grounding language to real-world robot actions [12], [13], handling ambiguous or user-specified instructions, and ensuring robust performance in unstructured environments. Our work builds upon these foundations by integrating LLM-based planning with open-vocabulary perception and legged robot mobility, aiming to address the limitations of previous approaches in long-horizon, open-world scenarios.

B. Open-Vocabulary Visual Perception

Open-vocabulary visual perception has evolved significantly from early fixed-class object detectors like YOLO [3], which are confined to recognizing predefined categories and struggled in open-world scenarios. Modern advancements have shifted toward models that leverage cross-modal representations, such as CLIP [14], which learns to align visual and language embeddings through contrastive pretraining on massive image-text pairs, enabling zero-shot object recognition by mapping visual features to arbitrary language descriptions. Subsequent works like Grounding DINO [15] and YOLO-world [16] have further enhanced the detection capability by integrating grounded pretraining to improve open-set detection accuracy. For robotic applications, real-time performance and robustness to dynamic camera motions, such as jittering caused by legged robot locomotion or temporary target occlusions, remain critical challenges [17]. Existing methods often fail to maintain detection stability in such scenarios, lacking the feedback mechanisms to adapt to the robot’s motion state or environmental changes. LOVON addresses these gaps by developing specialized preprocessing techniques to mitigate motion blur and ensure consistent visual input, while tightly integrating open-vocabulary detection with task planning and motion control for end-to-end execution in unstructured environments.

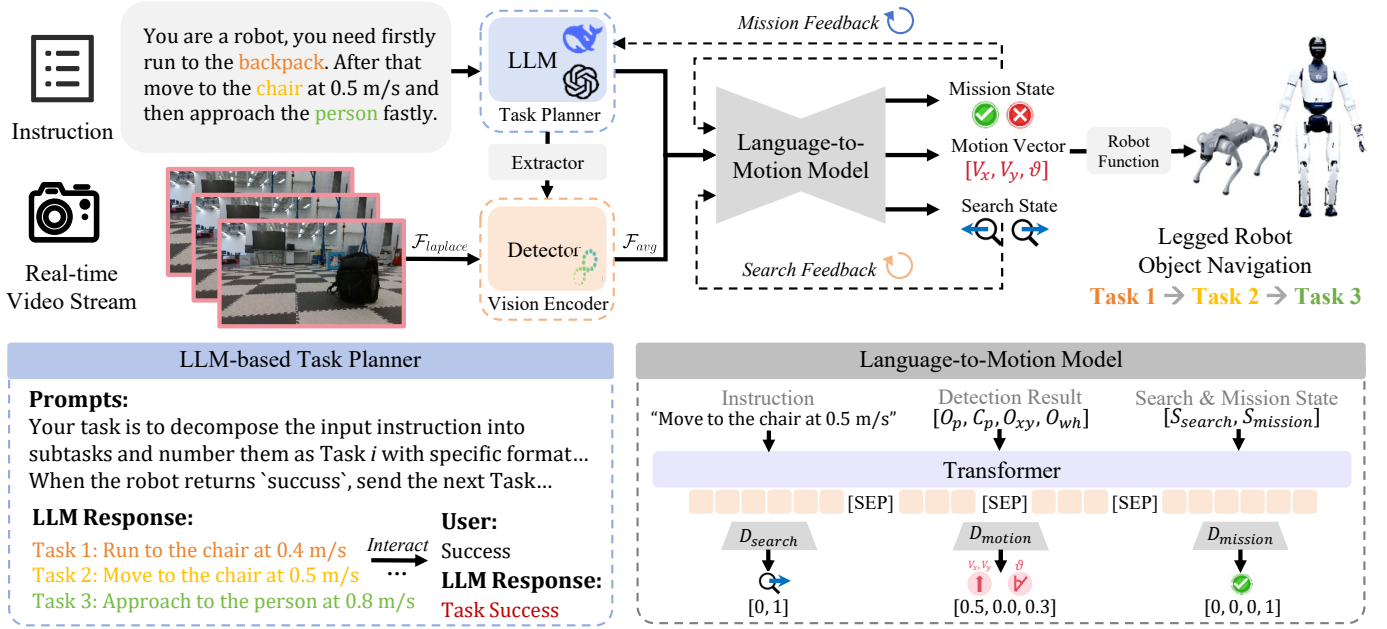


Fig. 2. **Overview of LOVON's pipeline.** The pipeline operates in three core stages: **1) Task Reconfiguration:** The LLM Task Planner transforms human-defined long-horizon tasks into structured basic mission instructions. **2) Video stream & Instruction object Processing:** The detection model processes the video stream, integrating a Laplace filter and instruction objects extracted via IOE. **3) Motion Generation & Feedback:** Mission instructions, target objects, bounding boxes, and robot states are input to the Language-to-Motion Model (L2MM). The L2MM generates the robot's control vector and provides feedback to both the LLM and itself, facilitating the progressive completion of all tasks.

C. Legged Robot Navigation and Long-Horizon Autonomy

Legged robot navigation has advanced from low-level mobility to complex tasks, but existing approaches often focus on single-task optimization [18]–[20] and lack integration of high-level planning for long-horizon missions [21]–[24]. While legged robots show superior terrain adaptability [25], their potential for executing sequential goals in unstructured environments remains underexplored, as traditional systems often separate perception from motion planning and struggle with dynamic target tracking. LOVON addresses this by unifying hierarchical task decomposition with real-time motion control. An LLM-based planner breaks down long-horizon tasks, while the L2MM maps instructions and visual feedback to dynamic motion vectors. This allows adaptive behaviors like switching to search states when targets are lost, validated across Unitree Go2, B2, and H1-2 platforms in diverse terrains. By integrating open-vocabulary perception with legged mobility, LOVON enables autonomous robots to navigate complex and dynamic environments.

III. PROBLEM FORMULATION

Task. The task involves operating in an arbitrary open-world environment, where the robot is required to perform long-horizon tasks to search for different targets. The long-horizon task T_l , is defined as a set of subtasks $T_l = \{T_i | T_1, T_2, \dots\}$, where each subtask corresponds to searching for a specific target O_i . The description of T_l is flexible (example in the bottom left of Fig. 2), allowing for varying mission objectives. The core challenge is for the robot to autonomously search for and identify different subgoals (targets), navigating toward them at different velocities based on mission instructions.

These subgoals can vary throughout the task, requiring the robot to adapt dynamically.

Goal. Our goal is to develop a dual-system model: (I) A high-level policy that can decompose complex task instructions T_l into individual subtasks with concrete instruction $I_{ins} = \{I_i | I_1, I_2, \dots\}$ and perform task planning; (II) A low-level policy that, based on specific subtask instructions I_i and video stream input I_{RGB} , can generate motion vectors $V_m \in R^3$ to achieve precise motion control. The model should be adaptable across various legged robots, ensuring versatility in real-world applications.

IV. METHODOLOGY

A. Overview

The pipeline of LOVON is illustrated in Fig. 2. Initially, the LLM reconfigures the human's long-horizon task into basic mission instructions. This decomposition ensures that high-level objectives (e.g., 'You are a robot, you need firstly run to the backpack. After that move to the chair at 0.5 m/s and then approach the person fastly.') are translated into actionable, step-by-step directives suitable for subsequent robotic processing. These directive mission instructions are then transmitted to an instruction object extractor (IOE) to accurately identify and extract key target objects from the instructions. The detection model processes the real-time captured video stream, with the input image preprocessed leveraging a Laplace filter which specifically addresses motion blur in the captured frames and the information extracted from IOE. Finally, the mission instruction, target object, bounding box, mission state, and search state are combined as inputs to the proposed L2MM, which generates the robot's control

vector and feedback states for both the LLM and L2MM to enable dynamic task adjustments.

B. Multimodal Input Processing

LOVON integrates two class of pretrained models: object detection model (e.g., [5], [15], [16], [26], [27]) for visual input processing and LLM (e.g., [1], [28], [29]) for long-horizon task management. The input to the LLM consists of the system description I_{sys} , the user's long-sequence task description T_l , and feedback from the L2MM O_f . Using this input, the LLM generates specific mission instructions I_i , enabling LOVON to execute long-sequence tasks by producing the necessary instructions to achieve the mission objectives:

$$I_{ins} = f_{LLM}(I_{sys}, T_l, O_f). \quad (1)$$

Then, our proposed IOE maps the instruction to the detection class. During deployment, L2MM operates at a high update rate during the process, yet the long-horizon task planning offered by LLM only requires execution once. While LLMs are technically capable of both high-level task planning and instruction-to-class mapping, using LLMs for the latter introduces unnecessary inefficiencies. It consumes excessive tokens, increasing inference cost, and prolongs latency. IOE is used to map task instructions to target detection classes in an efficient manner which not only significantly simplifies the LLM's prompt structure but also minimizes the LLM's inference frequency—ultimately achieving a more efficient and cost-effective solution for practical applications. IOE uses a two-layer transformer with a linear layer as head to predict the object class:

$$I_{object} = f_{IOE}(I_m) \in \mathbf{C}, \quad (2)$$

where $\mathbf{C}$ represents the set of classes that the detection model is capable of recognizing.

Regarding the visual processing, the object detection model takes an RGB image I_{RGB} and I_{object} as input and outputs the desired detection information as follows:

$$O_m, C_p, O_{xy}, O_{wh} = f_{det}(I_{RGB}, I_{object}). \quad (3)$$

We use the normalized format for the detection results, with the predicted object denoted as O_m , the confidence score as C_p , and the center position of the bounding box as $O_{xy} = [x_n, y_n]$. The width and height of the bounding box are represented as $O_{wh} = [w_n, h_n]$. Furthermore, the bounding boxes derived from the object detection model's output are processed using a moving average filter. This filtering operation smooths fluctuations in the bounding box coordinates, which in turn improves system stability—this improvement is especially useful for ensuring reliable performance in real-world deployments.

C. Laplacian Variance-Based Motion Blur Filtering

When a legged robot moves, mechanical fluctuations from its legs, joints, and body propagate to its mounted camera, causing motion blur in captured frames, as shown in Fig. 3. Several frames (e.g., 1, 2, 5, 7, 8) are particularly blurred

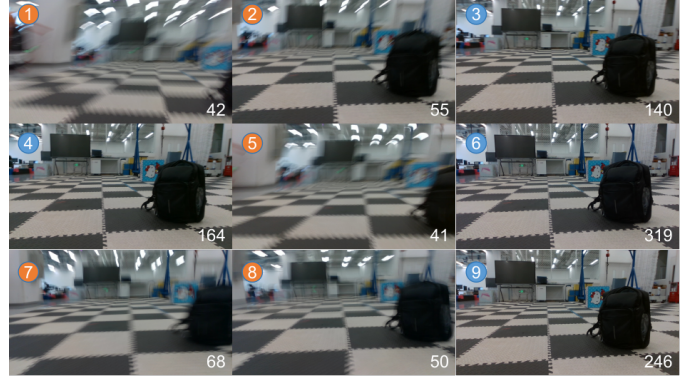


Fig. 3. **Image blurring phenomenon.** This figure shows the occurrence of image blurring in the robot view, which impacts the clarity and accuracy of the processed images. The number in the lower right corner of each image represents its Laplacian variance.

due to the robot's dynamic locomotion (e.g., step transitions, balance adjustments, or movement over rough terrain). For the downstream vision model, this blurred imagery affects its ability to extract visual features, locate objects, and align visual-language information, which are critical to the performance of the vision-language pipeline. To address this, we propose a Laplacian variance-based preprocessing technique for detecting and filtering motion-blurred frames. The method leverages the Laplacian operator's ability to highlight edge details: frames with low Laplacian variance (indicating blur) are identified and removed, while high-variance (clear) frames are retained. This step mitigates motion blur and movement-induced distortion, ensuring the object-detection-based vision-language pipeline receives high-fidelity inputs. In turn, it enhances the pipeline's robustness, reduces blur-related detection errors, and improves the reliability of the legged robot's vision-guided system.

In particular, we first convert the RGB frame I_{RGB} to a grayscale frame I_{gray} . Next, we apply the Laplacian operator to enhance high-frequency components, generating the Laplacian response. The variance of the Laplacian response is computed to assess the clarity of the frame. If the variance is below a threshold T_{blur} , the frame is classified as blurred and replaced by the last clear frame. For instance, when $T_{blur} = 100$, the motion-blurred frames labeled in orange (Fig. 3) are filtered out, preventing them from jeopardizing the whole program. The threshold T_{blur} is empirically calibrated for robot scenarios. The performance of this filtering method is discussed in Sec. VII-B.

D. Language-to-Motion Model

Our proposed L2MM is the core module responsible for predicting motion and providing feedback, enabling generalization beyond the limitations of static training instruction templates and robust performance on real-world unstructured inputs. Unlike rigid template-based methods that are confined to a fixed set of preprogrammed scenarios, the L2MM dynamically adapts to novel linguistic nuances and ambiguous command structures inherent in natural human instructions. It learns correlations between high-level instruction semantics

and low-level motion parameters, mapping abstract commands (e.g., imprecise description) to appropriate vector adjustments, which requires contextual understanding beyond explicit rule encoding.

L2MM is designed using an encoder-decoder architecture specifically designed to translate mixed inputs into robot motion commands. The encoder processes a structured sequence of diverse inputs that collectively capture task context, environmental objects perception, and robot state: these include the previous mission instruction I_{m0} for new mission awareness, the current mission instruction I_{m1} for immediate objectives, the predicted objects O_p , the predicted confidences C_p , the center positions O_{xy} , the width and height of the normalized bounding boxes O_{wh} , as well as the current mission state S_m and the current search state S_s track task progress and object-finding status. To unify these heterogeneous inputs, each component is converted to a string format and concatenated into a single sequence using [SEP] tokens as delimiters, forming the encoder input $I_{encoder} = \{I_{m0}, I_{m1}, O_p, C_p, O_{xy}, O_{wh}, S_m, S_s\}$. This sequence is processed by the encoder, which extracts contextual relationships between inputs and outputs a compact latent state l_e . This latent state is then passed to the decoder, which features three distinct, parallel heads—each specialized to handle a separate prediction task critical for generating coherent robot motion: **Motion Vector Head** D_{motion} . This head predicts the robot's motion vector V_m based on the latent state l_e . It is formulated as a sequence-to-vector problem, where the output is the control vector for the robot's movement:

$$V_m = D_{motion}(l_e). \quad (4)$$

Mission State Head $D_{mission}$. This head predicts the mission state S_m that is used as feedback for the encoder to adjust actions. The prediction is formulated as a sequence-to-number problem, where the output represents the current status of the mission. The prediction is given by:

$$S_m = D_{mission}(l_e). \quad (5)$$

Search State Head D_{search} . This head predicts the search state S_s , which indicates the robot's progress in searching for the target. It is also a sequence-to-number problem, where the output reflects the current state of the search:

$$S_s = D_{search}(l_e). \quad (6)$$

Each of these decoder heads uses a linear layer to process the latent state l_e and generate the respective outputs.

The final output of the model is a combination of the predictions from all three decoder heads:

$$O_{decoders} = \{V_m, S_m, S_s\}, \quad (7)$$

where $V_m = [v_x, v_y, \theta]$, v_x and v_y represent the robot's velocities along the x and y axes, and θ represents the angular velocity; S_m and S_s is the output search state and mission state, respectively. The mission state includes `success` (task completed successfully) and `running` (moving towards the detected object). The search state includes `searching_0` (searching by rotating left) and `searching_1` (searching

TABLE I
RELATIONSHIP BETWEEN STATES AND MOTION VECTOR. IN THE RUNNING STATE, v_x REPRESENTS THE MOVING SPEED OF x AXIS AND v_y REPRESENTS THE MOVING SPEED OF y AXIS. THE ROTATION θ_{corr} IS REGULATED BASED ON THE CENTER POSITION O_{xy} DURING EXECUTION TO CORRECT THE HEADING DIRECTION.

State S_m/S_s	Motion vector V_m
success	[0, 0, 0]
running	[v_x , v_y , θ_{corr}]
searching_0	[0, 0, -0.3]
searching_1	[0, 0, 0.3]

by rotating right). The relationship between states and their corresponding motion vectors is summarized in Table I.

This architecture enables the model to predict motion vectors, task states, and search states simultaneously, allowing the robot to not only control its motion accurately but also understand long task sequences and provide relevant feedback.

E. Loss Functions

The model is trained using different loss functions depending on the task:

Motion Vector Loss. For the motion vector head D_{motion} , we use the mean squared error loss with coefficient β to measure the difference between the predicted and actual motion vectors:

$$L_{MSE} = \frac{1}{N} \beta \sum_{i=1}^N (V_{m_{pred}}^i - V_{m_{true}}^i)^2. \quad (8)$$

Mission and Search State Loss. For the mission and search state heads $D_{mission}$ and D_{search} , we use cross-entropy loss to compare the predicted states with the ground truth labels:

$$L_{CE} = - \sum_{i=1}^N y_i \log(p_i), \quad (9)$$

where y_i is the true label, and p_i is the predicted probability for each class.

V. ROBOT FUNCTIONAL LOGIC FOR TASK EXECUTION

Once the model generates predictions, the robot follows its functional logic to execute tasks and adapt to environmental changes. The key functions that guide its behavior during task execution include:

- **Execute New Mission:** The robot compares the previous mission instruction with the current one. If they differ, the robot begins the new task.
- **Run to the Object:** Upon detecting the mission object, the robot navigates towards it based on the motion vector and the detection results.
- **Search for Lost Object:** If the robot loses track of the mission object, it automatically switches to a searching state and adjusts its motion to relocate the object.
- **Maintain the State:** The robot maintains its current state based on real-time visual inputs until a transition is triggered, ensuring consistent task execution.
- **Accomplish the Mission:** The robot monitors the mission object and, once O_{wh} is within a success threshold, it stops and transitions to the `success` state.

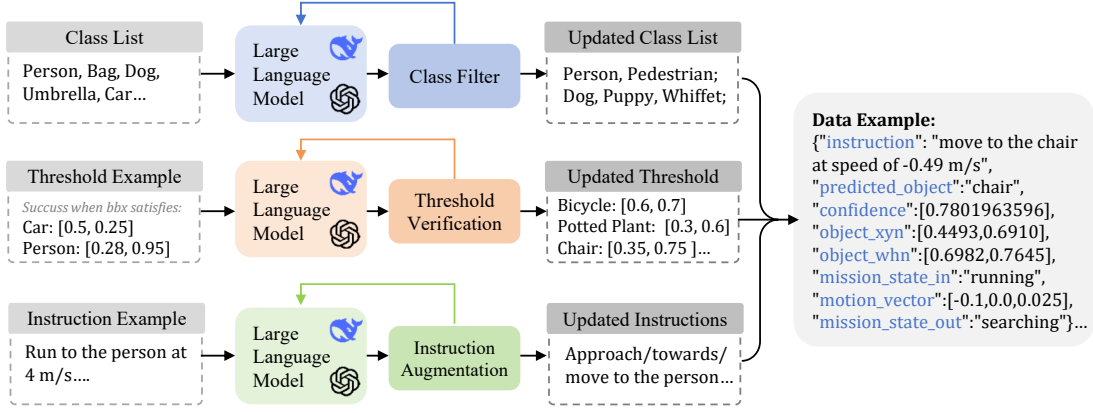


Fig. 4. **Illustration of the dataset preparation pipeline**, including three key functional steps and an iterative optimization loop. The three steps are: 1) Detection Class Synonym Expansion (LLM generates object class synonyms to enrich categories and model generalization); 2) Instruction Variation (LLM creates instruction paraphrases to support diverse sentence structures and maintain core info, which improves adaptability to real-world diverse inputs); 3) Threshold Generation for Object Categories (it defines initial detection thresholds, then uses LLM to adapt them for other categories to address different object sizes). Complementing these steps, an LLM-driven feedback loop iteratively refines instruction templates, reducing dataset redundancy and enhancing its diversity over time.

These functional rules are used to determine the operational state of the robot for data labeling, ensuring autonomous navigation, task adaptation, and robust task completion.

VI. DATASET PREPARATION

As illustrated in Fig. 4, the dataset preparation pipeline consists of three main components:

Detection Class Synonym Expansion. We use an LLM to generate synonyms for the predefined object classes, enriching the object categories and improving the model’s ability to generalize across different object descriptions.

Instruction Variation. To enhance the language module, we use the LLM to generate paraphrases of instructions, allowing the model to process diverse sentence structures while preserving core information, improving its adaptability.

Threshold Generation for Object Categories. We define success thresholds for object detection based on initial examples, then use the LLM to adapt these thresholds for other categories, ensuring the model handles different object sizes.

During dataset preparation, the instruction templates are fed back into the LLM to iteratively refine the dataset, avoiding redundancy and improving the dataset’s diversity over time. After obtaining the instruction templates, we randomly generate the target object from detection class synonyms, speed values, object bounding box, etc., and substitute them into the templates. Then, the robot states are determined based on the values in the templates ruled by the robot function logic, and the ground-truth V_m is obtained according to Table I to produce the final training data.

The dataset preparation process is fast and easy to expand. It takes less than 15 minutes to process 1 million data with CPU Intel i9-12900KF.

VII. EXPERIMENT

A. Experiment Setup

Model Details. For object detection, L2MM is designed to take inputs in a general string format, which can be directly

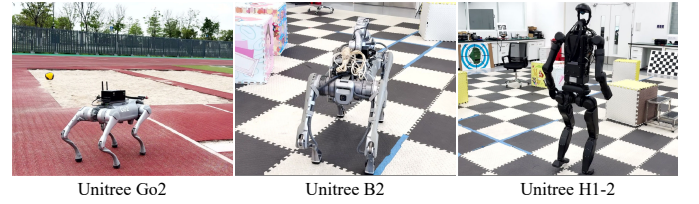


Fig. 5. **LOVON with multi-configurations.** LOVON can be seamlessly adapted to any legged robot for precise object navigation.

obtained from normal detection models. Thus, LOVON is independent of detection models. We employ the recently developed detection model, YOLO-world [16], which offers open-vocabulary detections with high performance and lightweight architecture. As the task planner and data generation assistant, we utilize DeepSeek R1 [28]. L2MM is a transformer-based model with feature dimension 256, 4 layers, and 8 attention heads, a feedforward dimension of 1024, and 1 linear layer as decoder for each head. IOE shares the same general architecture, with a reduced feature dimension size of 64, 2 transformer layers with 4 attention heads, and a feedforward dimension of 256.

Training Settings. Our dataset consists of 1 million samples, which are divided into training and testing sets in a 4:1 ratio. We use the NVIDIA RTX 3080 Ti GPU for training. The L2MM model is trained with a dropout rate of 0.1, a learning rate of 10^{-4} , a batch size of 512, a max-sequence length of 64, and a motion loss coefficient β set to 10. It is trained 25 epochs using the AdamW optimizer. The total training time is approximately 1 hour. Similarly, the IOE model is trained with the same training settings by approximately 30 minutes.

Robot Settings. LOVON is versatile and can be applied to various legged robots. In our experiments, we evaluate three representative models: Unitree Go2, B2, Unitree H1-2, as shown in Fig. 5. The L2MM velocities V_m are directly fed to Unitree’s API for low-level velocity tracking. For the computing platform, we utilize the Jetson Orin, while the visual platform consists of the robots’ built-in cameras and the Realsense D435i camera.

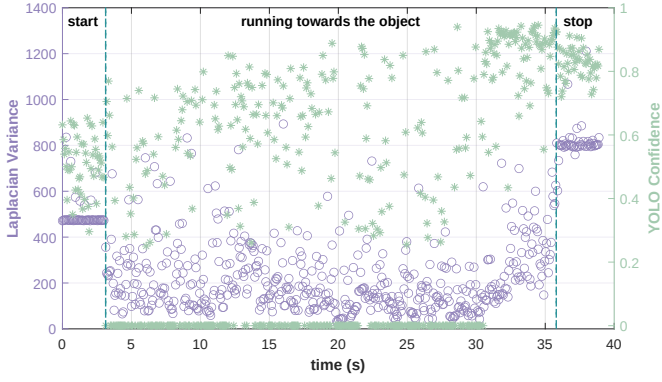


Fig. 6. Visualization of the Laplacian variance and the confidence of the detection object (0.3 m/s towards the chair case). In the start and stop periods, the robot stays stationary. During the running stage, the robot is commanded to approach a backpack/chair/person at fixed speeds of 0.3/0.5/0.7 m/s, with the front camera updating at 15 Hz and keeping the target object in view. After capturing frames, we calculate each frame’s Laplacian variance and feed the frame into an object detection model to get predicted confidence scores.

B. Performance of Motion-Blurred Frame Filtering

To investigate the impact of motion blur on object detection, we conduct experiments to analyze the relationship between Laplacian variance and detection confidence. We command the robot to approach a backpack, chair, or person at fixed speeds of 0.3 m/s, 0.5 m/s and 0.7 m/s. The camera updates at ~ 15 Hz. During the running process, we ensure the target objects is visible by the front camera. After capturing the frames, we compute the Laplacian variance for each frame and input it into the object detection model to obtain predicted confidence scores. As shown in Fig. 6, the Laplacian variance and YOLO confidence fluctuates significantly, particularly during the running process. Additionally, we can observe that low detection confidence often coincide with a low Laplacian variance, and there is a close relationship between them. In the running phase, even though the target object remains continuously within the field of view of the front camera, the detection model still frequently fails to recognize it (resulting in zero confidences). This phenomenon can be attributed to the motion blur introduced by the robot’s movement, which degrades the image quality and impairs the detection model’s capability to accurately identify and localize the target object.

To address this, we adopt our proposed motion-blurred frame filtering method. By setting a blur threshold introduced in Sec. IV-C, we can filter out frames with excessive motion blur utilizing the Laplacian variance. The impact of varying blur thresholds is investigated in Fig. 7, where we observe that higher thresholds result in a higher qualified frame ratio. However, this trend has a practical limit: setting the threshold too high undermines the filter’s purpose by allowing marginally blurred frames to pass through, and in extreme cases, it can even lead to the unnecessary rejection of valid, sharp frames. This occurs because an overly high threshold misclassifies subtle but acceptable sharpness variations as ‘blur’, excluding frames that would otherwise contribute to accurate object detection. After testing, we set the threshold to $T_{blur} = 150$,

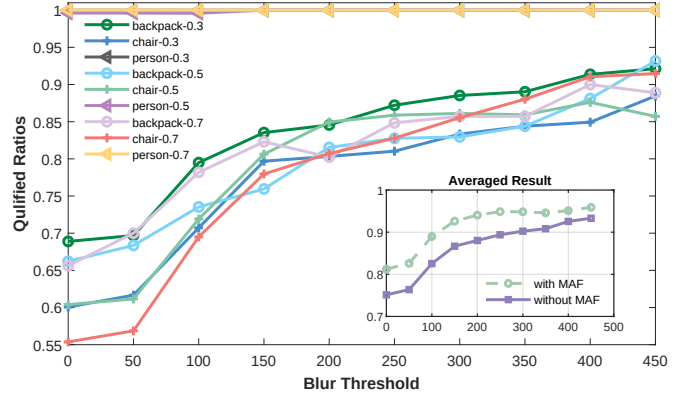


Fig. 7. Thresholds and qualified ratios of different objects at different speeds. As shown in the main figure, increasing the blur threshold directly leads to a higher qualified frame ratio, but setting the threshold too high can lead to unnecessary rejection of valid frames. As shown in the subplot, our filter method with MAF increases the qualified frame rate by 25% with $T_{blur} = 150$ and MAF window length of 3.

which improves the qualified frame ratio by approximately 15% for all sets. We then integrate this filtering method into our object detection pipeline. Blurred frames are excluded and replaced with the last qualified frame, and the detection confidence is smoothed using a moving average filter (MAF). For this MAF, we set the window length to 3. As shown in the subplot of Fig. 7, this integration leads to an overall 25% increase in the qualified frame rate, demonstrating the effectiveness of our motion-blurred frame filtering method.

C. Evaluation in Simulation Environments

Benchmark and Evaluation Metric. In our evaluation, we follow the setup from previous works [30] under the Gym-Unreal benchmark [6] (as shown in Fig. 8), where the maximum episode length is 500 steps. The visible region of the tracker is defined as a 90-degree fan-shaped sector with a radius of 750 cm. We evaluate performance using two metrics: 1) Episode Length (EL), which measures the average duration of episodes over 100 trials, reflecting the tracker’s long-term performance; 2) Success Rate (SR), which calculates the percentage of successful episodes across 100 trials, indicating the model’s robustness.

Baselines. 1) **DiMP** uses a pre-trained video tracker for bounding box and tunes a PID controller to control the tracker. 2) **SARL** trains an end-to-end Conv-LSTM network via online reinforcement learning (RL). 3) **AD-VAT** trains the RL-based tracker using an asymmetric dueling mechanism with a learnable adversarial target to improve the robustness of the end-to-end tracker. 4) **AD-VAT+** applies a two-stage training strategy to learn in complex environments with obstacles. 5) **TS** uses a pose-based teacher to provide online suggestions to the vision-based student while interacting with environments. 6) **RSPT** utilizes RGB-D and combines a video tracker to construct a structure-aware representation and train the policy via RL. 7) **EVT** trains a pre-trained visual foundation model with offline RL. 8) **TrackVLA** employs an LLM for recognition and an anchor-based diffusion model for trajectory planning.

Performance Comparisons. As shown in Table II, our LOVON outperforms several baseline approaches, achieving

TABLE II

QUANTITATIVE RESULTS COMPARED WITH BASELINES IN GYM-UNREAL ENVIRONMENTS. THE NUMBERS OF EACH CELL REPRESENT THE AVERAGE EL AND SR. *DiMP USES A PRETRAINED VIDEO TRACKER WHICH DOES NOT NEED ADDITIONAL TRAINING TIME. ALL THE OTHER BASELINES ARE RL-BASED EXCEPT FOR TRACKVLA, WHICH FOLLOWS THE STANDARD PRACTICE IN VISION-LANGUAGE MODELING. **MEAN-1:** MEAN VALUE ACROSS DIFFERENT SCENES FOR PERSON TRACKING. **MEAN-2:** MEAN VALUE ACROSS DIFFERENT ANIMAL TRACKING.

Methods	Training time	ParkingLot	UrbanCity	UrbanRoad	SnowVillage	Mean-1	Horse	Dog	Sheep	Mean-2
DiMP [31]	0 hours*	327/0.48	401/0.66	308/0.33	301/0.43	334.25/0.48	-/-	-/-	-/-	-/-
SARL [32]	24 hours	301/0.22	471/0.86	378/0.48	318/0.31	367.00/0.47	-/-	-/-	-/-	-/-
AD-VAT [33]	12 hours	302/0.20	484/0.88	429/0.60	364/0.44	394.75/0.53	-/-	-/-	-/-	-/-
AD-VAT+ [34]	24 hours	439/0.60	497/0.94	471/0.80	365/0.44	443.00/0.70	-/-	-/-	-/-	-/-
TS [35]	26 hours	472/0.89	496/0.94	480/0.84	424/0.63	468.00/0.83	-/-	-/-	-/-	-/-
RSPT [36]	12 hours	480/0.80	500/1.00	500/1.00	410/0.80	472.50/0.90	-/-	-/-	-/-	-/-
EVT [30]	1 hours	484/0.92	500/1.00	496/0.96	471/0.87	487.75/0.94	500/1.00	469/0.90	471/0.93	480/0.94
TrackVLA [23]	360 hours	500/1.00	500/1.00	500/1.00	500/1.00	500.00/1.00	500/1.00	500/1.00	500/1.00	500/1.00
LOVON (Ours)	1.5 hours	500/1.00	500/1.00	499/0.99	500/1.00	499.75/1.00	500/1.00	500/1.00	500/1.00	500/1.00

TABLE III
MODEL SIZE

Model Name	Size	Notes
DiMP [31]	DiMP50 (364 MB)	The DiMP50 is based on ResNet-50.
AD-VAT+/AD-VAT [33]	AD-VAT-main (3.2 MB)	It gets processed perceptions from simulator.
TS [35]	TS-VAT-main (3.2 MB)	It follows the same way as AD-VAT with extractors.
RSPT	RSPT-main (3.1 MB)	It follows a similar way as AD-VAT.
EVT [36]	DEVA (277 MB), GroundingDino (694 MB), EVT-main (2.7 MB)	It uses pretrained models for visual inputs.
TrackVLA [23]	LLM Backbone + Heads (13.5GB+)	It is finetuned based on the Vicuna-7B [37].
LOVON (Ours)	YOLO-World (141 MB), IOE (591 KB), L2MM (13.3 MB)	We use pretrained models for detection.

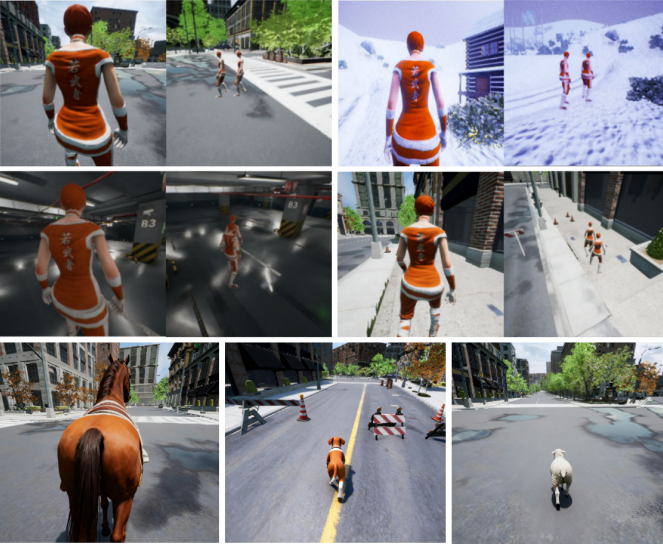


Fig. 8. **Simulation evaluation.** We conduct extensive experiments in Gym-Unreal to comprehensively test the model’s robustness and generalization. The evaluation covers two key dimensions: 4 distinct scenes (UrbanCity, SnowVillage, ParkingLot, UrbanRoad) and 3 additional target objects (Horse, Dog, Sheep) to assess performance beyond common benchmark objects.

a perfect SR of 1.00 across most environments, including ParkingLot, UrbanCity, SnowVillage, Horse, Dog and Sheep series. Compared to strong baseline-model EVT [30], LOVON demonstrates superior tracking performance, e.g., 500/1.00 vs. 484/0.92 in ParkingLot, as a result of its integrated searching capability. Moreover, LOVON excels EVT in all animal tracking situations. As the animals have varying target sizes, gait patterns, and environmental interactions, the results relect LOVON’s ability to adapt to real-world challenges distinct from person-tracking. Fig. 9 shows some success cases



Fig. 9. **Tracking failure cases.** The baseline models (e.g., EVT, RSPT) perform better in the UrbanCity scene than in other scenes. Failure cases in the ParkingLot scene are mainly due to losing the target person. LOVON’s integrated searching capability boosts its tracking performance in cluttered scenes, where alternatives often fail to recover tracking after target loss.

and failure cases of the tracking process in simulation. All the baseline models (e.g., EVT, RSPT) perform well in the UrbanCity scene because the space is large and the target person is not easy to lose. While most of the baseline models do not perform well in scenes like ParkingLot (Table II). These models often fail when the target person turns fast and the tracker loses the target’s views. LOVON can search from the environment when the target person turns fast and the tracker loses the target’s view, and continue the tracking successfully. Even when compared to the state-of-the-art TrackVLA [23], which achieves 1.00 SR but requires 360 hours of training, LOVON stands out with an efficient training time of just 1.5 hours, offering both high accuracy and significant efficiency.

Table III presents a comparison of the model sizes of some selected baseline methods, highlighting key differences in architectural efficiency. LOVON stands out for achieving



Fig. 10. **Environment adaptation.** LOVON excels in open-world object navigation, effectively adapting to a wide range of objects and environments and is robust to unexpected disruptions.

the smallest overall model size apart from the models (e.g., AD-VAT) that take the advantage of the simulator to get the processed perception. This compactness of LOVON is particularly significant as it directly translates to practical advantages for deployment on resource-constrained legged robots, where memory and computational limits are critical constraints. Notably, LOVON’s efficiency does not come at the cost of prohibitive training times as the training components (IOE and L2MM) are relatively small. When compared to established baselines such as EVA and TrackVLN—models known for their robust performance but larger footprints, LOVON maintains competitive performance.

D. Evaluation on Real-World Experiments

In real-world evaluations where environments are unstructured, targets are dynamic, and unexpected disruptions are common, LOVON demonstrates exceptional performance in four key areas that directly address the practical demands of legged robot operation:

- (1) It excels in *open-world adaptation* (Fig. 10), allowing the robot to handle a wide range of objects commonly encountered in daily life, including large objects like cars, medium-sized ones like people, and small items such as bags. This enables LOVON to seamlessly interact with various objects, regardless of their size or type, in unfamiliar environments.
- (2) LOVON achieves *multi-goal tracking* through our LLM planner, enabling long-horizon object navigation. (The upper left corner of the Fig. 1). This capability allows the robot to efficiently track multiple objects over extended periods, even when the environment becomes more complex.
- (3) LOVON excels in *dynamic tracking*, successfully following moving objects in dynamic environments, much like walking a dog. We test this feature on flat roads, spiral stairs, and wild grass, and the robot reliably completes the task in these challenging conditions.
- (4) LOVON is *robust to disturbances*. If the targeted object is displaced or if the robot itself is disturbed (such as being kicked), the robot quickly re-localizes and continues its

TABLE IV
ABLATION STUDY ON MODEL PARAMETERS. WE TRAIN ALL THE MODELS FOR 50 EPOCHS AND SELECT THE MODEL WITH THE LOWEST VALIDATION LOSS FOR EVALUATION.

Model Size	N_{ds} (m)	β	[SEP]	$\sigma_v (\times 10^{-4}) \downarrow$	$\epsilon_v (\times 10^{-2}) \downarrow$
LOVON-Small (0.47M)	0.4	10	✓	3.74	5.74
LOVON-Large (25.51M)	0.4	10	✓	0.43	0.23
LOVON-Base (3.30M)	0.2	10	✓	3.73	0.43
	0.8	10	✓	0.40	1.49
	0.4	1	✓	4.59	7.88
	0.4	20	✓	Fail	Fail
	0.4	10	×	1.02	29.5
LOVON-Base (Best)	0.4	10	✓	0.38	1.80

TABLE V
ABLATION STUDY ON PROPOSED METHODS. N_s : NUMBER OF SEARCHING; T_s : SEARCH TIME (S)

Object	States	Filter	4 m		6 m	
			$N_s \downarrow$	$T_s \downarrow$	$N_s \downarrow$	$T_s \downarrow$
Backpack	3	×	4.05	100.00	5.95	178.00
	4	×	1.60	45.91	2.25	83.09
	4	✓	1.00	22.25	1.00	32.57
Chair	3	×	2.05	43.50	4.25	110.00
	4	×	1.05	26.29	1.35	45.34
	4	✓	1.00	21.73	1.00	30.55
Person	3	×	1.00	19.80	1.00	28.00
	4	×	1.00	17.66	1.00	26.24
	4	✓	1.00	17.47	1.00	26.51

search. In one test, when we move the chair and kicked the robot, LOVON still enables the robot to approach the chair. Additionally, in a playground with sandy terrain, even when a sports ball is kicked away, the robot completes its task without difficulty.

E. Ablation Study

Ablation on the Model Parameters.

We investigate the impact of model size, dataset size N_{ds} , motion loss weight β , and the inclusion of special tokens. The model’s performance is evaluated by comparing the standard deviation σ_v and the average speed bias ϵ_v compared to the speed given in the instruction (0.40 m/s). As shown in Table IV, the base-size model shows good performance, while smaller models exhibit higher σ_v and ϵ_v , indicating they cannot effectively capture the required information. Larger models generally perform better, but the improvement in speed tracking is marginal despite the model size being much larger. In addition, models trained on smaller datasets show lower ϵ_v but higher σ_v , reflecting instability. Larger datasets improve performance but do not significantly impact real-world applicability. The motion loss weight β plays a crucial role. A smaller $\beta=1$ results poorly, as the model gives insufficient attention to motion loss. Conversely, a larger $\beta=20$ leads to undervaluation of state loss, causing inaccurate state inference. Finally, the inclusion of the special token [SEP] is essential. The model trained without this token struggles to follow the given speed, as [SEP] helps differentiate between various input components, especially language.

Ablation on the Filter Method and the Number of States.

We evaluate the impact of the number of searching states and the frame-filtering technique on the efficiency of object navigation when the target is lost. Three system configurations

are tested: Case 1, with three states and no frame filtering; Case 2, with four states but no frame filtering; and Case 3, with four states and frame filtering. Experiments are conducted on the Go2 robot with three objects (backpack, chair, and person) at distances of 4m and 6m.

As shown in Table V, LOVON excels in seeking the person, which is easily detected, while the backpack, which is harder to track, requires more effort. In Case 1, the robot performs inefficiently, often losing the object and requiring significant time due to motion blur. Adding a mission state in Case 2 improves efficiency by enabling the robot to search in both directions, though object loss and shaking still occur, leading to a higher number of searching circles. In Case 3, with four states and the frame-filtering technique, LOVON achieves optimal efficiency. It reduces N_s to 1 for both the backpack and chair, matching the performance with the person. Furthermore, the time to reach the target is reduced by 5 and 2 times for the backpack and chair, respectively, compared to Case 1, approaching the performance observed for the person.

VIII. CONCLUSION

In conclusion, we introduce LOVON, a model designed to tackle three core challenges in legged robot vision-guided tasks: executing long-horizon missions, adapting to unstructured real-world environments, and achieving state-of-the-art object tracking performance. To enhance its reliability in practical scenarios, we integrated two key preprocessing and optimization modules: a Laplacian-variance-based frame filter, which effectively eliminates motion-blurred frames that degrade detection accuracy, and a moving average confidence filter, which smooths fluctuating object detection scores to stabilize tracking. Beyond these enhancements, LOVON leverages the high-level reasoning capabilities of LLMs to decompose complex long-horizon tasks (e.g., multi-object sequential navigation) into actionable steps, ensuring consistent progress even in extended missions. We validated LOVON's performance through extensive evaluations: simulation tests in the Gym-Unreal environment (covering diverse scenes like UrbanCity and SnowVillage) and real-world experiments (spanning indoor spaces, outdoor terrain, and dynamic scenarios) both confirm its robustness and effectiveness across open-world, unstructured conditions.

Despite these strengths, LOVON currently has limitations in contextual understanding as its focus remains primarily on object semantics and motion tracking, with limited ability to interpret broader scene context (e.g., inferring 'a chair in a crowded living room' vs. 'a chair in an empty garage'). In future works, we aim to improve contextual understanding by incorporating more powerful vision-language foundation models (e.g., Llama [29], [38]), which can fuse visual features with contextual language understanding to enrich scene awareness. Additionally, we plan to incorporate advanced perception techniques (such as depth estimation and semantic segmentation) to expand LOVON's applicability to more diverse practical tasks like object manipulation and human-robot collaborative interactions. These improvements will further bridge the gap between LOVON's current capabilities and the demands of fully autonomous, real-world legged robot operation.

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